

1. Explain MITRE ATT&CK

MITRE ATT&CK stands for **Adversarial Tactics, Techniques, and Common Knowledge**. It is a cybersecurity framework developed by the MITRE Corporation to help security professionals understand how cyber attackers operate.

Instead of focusing only on malware or viruses, the MITRE ATT&CK framework explains the different steps an attacker takes to compromise a system. It is based on real-world cyberattack observations and is widely used by Security Operations Centers (SOCs), incident responders, threat hunters, and security analysts.

Organizations use this framework to improve their security by identifying weaknesses, creating better detection rules, and responding to attacks more effectively.

Why is MITRE ATT&CK important?

- It helps security teams understand attacker behavior.
- It improves threat detection and monitoring.
- It supports incident response and threat hunting.
- It helps organizations strengthen their security controls.
- It provides a common language for cybersecurity professionals around the world.

2. Define Tactics and Techniques

Tactics

A **tactic** represents the attacker's objective or goal during a particular stage of an attack. In simple words, it answers the question:

"What is the attacker trying to achieve?"

Some common tactics include:

- Initial Access
- Execution
- Persistence
- Privilege Escalation
- Credential Access
- Discovery
- Lateral Movement
- Command and Control
- Exfiltration
- Impact

For example, if an attacker wants to steal passwords, the tactic is **Credential Access**.

Techniques

A **technique** explains the method used by the attacker to achieve a tactic. It answers the question:

"How is the attacker achieving the goal?"

For example:

- **Tactic:** Initial Access
- **Technique:** Phishing Email

In this case, the attacker gains access to the victim's computer by sending a fake email containing a malicious attachment or link.

Difference between Tactics and Techniques

Tactics	Techniques
Describe the attacker's goal	Describe how the goal is achieved
High-level objective	Specific method used
Example: Initial Access	Example: Spear Phishing

3. Analyze a Sample Attack

Scenario: Phishing Attack Leading to Ransomware

Imagine an employee receives an email that appears to come from their manager. The email contains an attachment and asks the employee to open it urgently.

Step 1 – Initial Access

The employee opens the attachment, which contains malicious code.

- **Tactic:** Initial Access
- **Technique:** Spear Phishing Attachment

The attacker successfully gains access to the computer.

Step 2 – Execution

Once the attachment is opened, a hidden script starts running in the background and downloads malware onto the system.

- **Tactic:** Execution
- **Technique:** User Execution

Step 3 – Persistence

The malware creates a registry entry so that it automatically starts every time the computer is turned on.

- **Tactic:** Persistence
- **Technique:** Registry Run Keys

Step 4 – Credential Access

The attacker attempts to steal saved usernames and passwords from the infected computer.

- **Tactic:** Credential Access
- **Technique:** Credential Dumping

Step 5 – Lateral Movement

Using the stolen credentials, the attacker moves to other computers within the same network.

- **Tactic:** Lateral Movement
- **Technique:** Remote Services

Step 6 – Command and Control

The infected computer secretly communicates with the attacker's remote server over the internet to receive further instructions.

- **Tactic:** Command and Control
- **Technique:** Application Layer Protocol

Step 7 – Impact

Finally, the attacker encrypts important files on the organization's computers and demands a ransom payment to restore access.

- **Tactic:** Impact
- **Technique:** Data Encrypted for Impact